

48) Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x \forall y (P(x) \vee Q(y))$, where all quantifiers have the same nonempty domain, are logically equivalent. (The new variable y is used to combine the quantifications correctly).

Soln: Let, $\forall x P(x) \vee \forall x Q(x)$ be False. By defn. of OR, $\forall x P(x)$ is False and $\forall x Q(x)$ is False. By defn. of universal quantifier, there exists an element a in the domain such that $P(a)$ is False. By defn. of universal quantifier, there exists an element b in the domain such that $Q(b)$ is False. By defn. of OR, $P(a) \vee Q(b)$ is False. By defn. of universal quantifier, $\forall y (P(a) \vee Q(y))$ is False. By defn. of universal quantifier, $\forall x \forall y (P(x) \vee Q(y))$ is False.

Let $\forall x \forall y (P(x) \vee Q(y))$ is False. By defn. of universal quantifier, there exists an element a in the domain s.t. $\forall y (P(a) \vee Q(y))$ is False. By defn. of universal quantifier, there exists an element b in the domain s.t. $(P(a) \vee Q(b))$ is False. By defn. of OR, $P(a)$ is False and $Q(b)$ is False. By defn. of universal quantifier, $\forall x P(x)$ is False. By defn. of universal quantifier, $\forall x Q(x)$ is False. By defn. of OR, $\forall x P(x) \vee \forall x Q(x)$ is False.

$$\therefore \forall x P(x) \vee \forall x Q(x) \equiv \forall x \forall y (P(x) \vee Q(y)) \quad \square$$

49) a) Show that $\forall x P(x) \wedge \exists x Q(x)$ is logically equivalent to $\forall x \exists y (P(x) \wedge Q(y))$, where all quantifiers have the same non-empty domain.

Soln: Let, $\forall x P(x) \wedge \exists x Q(x)$ be True. By defn. of AND, $\forall x P(x)$ is True and $\exists x Q(x)$ is True. By defn. of universal quantifier, for all elements a in the domain, $P(a)$ is True. By defn. of existential quantifier, there exists an element b in the domain s.t. $Q(b)$ is True. By defn. of AND, $P(a) \wedge Q(b)$ is True. By defn. of existential quantifier, $\exists y (P(a) \wedge Q(y))$ is True.